

Evaluation of Muscle Synergies

Analysis of muscle synergies in subjects during treadmill walking and running

Gruppo 23

Fioretti Michele, Maffei Simone, Tarditi Valerio, Traverso Matteo Giovanni

Abstract — This study focuses on the characteristics of muscle synergies during treadmill walking and running tasks. The dataset includes surface EMG (sEMG) recordings of 13 ipsilateral muscles from 30 healthy adult participants, balanced for gender. Data preprocessing involved filtering, normalization, and artifact removal, utilizing custom algorithms for signal refinement. Synergies were extracted using non-negative matrix factorization (NNMF), with statistical consistency ensured by sorting synergies via the Hungarian algorithm.

Results highlighted four primary synergies corresponding to the four distinct biomechanical functions across locomotion types. Statistical analysis revealed significant differences between walking and running synergies, particularly in weight acceptance phases, while minimal differences were observed between genders. These findings suggest that locomotion type strongly influences muscle synergy patterns, whereas gender does not.

This study underscores the distinct biomechanical demands of walking versus running and validates the use of synergy analysis for understanding neural control of movement.

Keywords — muscle synergies, EMG, treadmill locomotion, NNMF, biomechanics.

I. INTRODUCTION

This work is focused on the analysis of the muscle synergies characteristics of two different treadmill locomotion types. The dataset consisted of EMG signals acquired from 30 subjects, equally divided in 15 females and 15 males. They were all healthy young adults, of different ages and different body types.

They were involved in 2 experimental tasks on a treadmill: during the first one the subjects were requested to walk at a speed of 1.4 m/s while during the second one to run at 2.8 m/s, as published in [1].

From this group of subjects, it was recorded the sEMG activity of 13 muscles, all of them ipsilaterally placed in the right side of the subject.

The aim of the work was to study the muscle synergies extracted for walking and running tasks, evaluating the possible correlations and differences with respect to sex or locomotion type.

II. MATERIALS AND METHODS

For EMG acquisition it was used a 16-channel wireless bipolar EMG system with a sampling frequency of 2kHz. This EMG acquisition system was used to get the acquisition of 13 muscles, which are gluteus MEdius (ME), gluteus MAximus (MA), Tensor Fasciae Latae (FL), Rectus Femoris (RF), Vastus Medialis (VM), Vastus Lateralis (VL), SemiTendinosus (ST), Biceps Femoris (BF), Tibialis Anterior (TA), Peroneus Longus (PL), Gastrocnemius Medialis (GM) Gastrocnemius Lateralis (GL), and Soleus (SOL). The muscles are the same studied in article [2].

To ensure a more consistent evaluation of muscle synergies, our first goal was the division of the original dataset into subgroups having the highest number of similarities as possible. To do that, a first separation based on subject genders was performed.

Moreover, we evaluated the BMI coefficient of each subject to discover any possible outliers in the dataset. Specifically, we calculated the BMI as the ratio between the subject's weight and his squared height value:

$$BMI = Weight / (Height^2) \quad (1)$$

Later, we also had to remove many subjects for both men and women subgroups, since they were causing problems with synergies, or they had EMG signals greatly corrupted by noise.

After this first data cleaning, we obtained a dataset divided into four subgroups as much comparable and coherent as possible:

- Subgroup 1: Men Walk (MW).
- Subgroup 2: Men Run (MR).
- Subgroup 3: Women Walk (WW).
- Subgroup 4: Women Run (WR).

After that we continued with the resampling and filtering phase for each subgroup. We decided to consider the single gait (or run) cycle split evenly between stance and swing phase resampling them at 500 samples and not at 100 each as described in [3]; this decision was taken according to our will to give the cycles a better representation, avoiding the loss of the majority of the variability among the subjects, but still maintaining an equal distinction between stance and swing in order to give them the same importance.

Subsequently, the EMG signals were high-pass filtered at the frequency of 50 Hz to eliminate slow motion artifacts, and then rectified and smoothed with a low-pass filter at 20 Hz using a sixth-order Butterworth filter. EMG patterns were normalized to the respective maximum amplitude.

Before getting the envelope, thanks to the low pass filter, we analyzed each subject's EMG signal to detect residual motion artifacts or any previously undetected problems. For this reason, we created a function called *emgpolish.m* that calculates a threshold as a multiple of the standard deviation and then applies it to the entire signal. The portions of the signal higher than it are forced by the function to NaN values and then replaced via a linear interpolation. It was possible to use this technique because we were replacing a few samples (about 10) on a single cycle consisting of 1000 samples, therefore we do not induce a significant change in the morphology of the signal with a consequent loss of information.

We extracted muscle synergies for each subject separately for all subgroups, using NNMF algorithm. The cycles given to the NNMF were established a priori, so that they weren't affected by the initial or the final change of velocity of the treadmill, for this reason we analysed the central 30 cycles (from the 16th to the 45th).

Next, we had to find the optimal number of synergies, to do that we used a threshold for the variance accounted for (VAF) fixed at the value of 90%. Specifically, the first synergy that was described by a VAF higher than the threshold was considered the optimal one. Moreover, to make the analysis more consistent, we controlled with a second threshold of 75% on every single muscle VAF, that there were no muscle channels highly corrupted by noise, in that case we came back and furtherly removed artifacts with *emgpolish.m* function. Finally, all these considerations led us to identify the correct number of muscle synergies for all the subgroups, obtained by extracting the mode of all optimal synergies found.

At this point, for each subgroup, it was necessary to reorder the synergies obtained for each patient to make the comparison between them easier. We opted for an association approach called Hungarian Algorithm, also known as the assignment problem or Kuhn-Munkres algorithm [4]. This is a combinatorial optimization algorithm used to solve assignment problems and was implemented with the support of a MATLAB function *munkres.m*, able to build a cost matrix defining the best associations between the several subject's synergies.

Once we obtained and ordered the synergies for each subgroup, we extracted some main parameters to analyse them in a more qualitative way and to be able to compare the results of the various subgroups with each other. Regarding time profiles, we extracted the mean value and its standard error, to estimate the neural activity related to the specific biomechanical function of that specific synergy. Regarding the muscular contributions, we calculated the average value of percentage activation of each of the 13 muscles taken into consideration and their relative standard error.

To make sure that these were some statistically significant parameters, we made a bilateral control with the 'p-value' t-test [5]. In this test we compared the values with 5% ($\alpha = 0.05$) as a significance level, to realize a cut-off for statistical significance.

Here you can see a flow-chart showing the steps followed during our work. (Fig. 1)

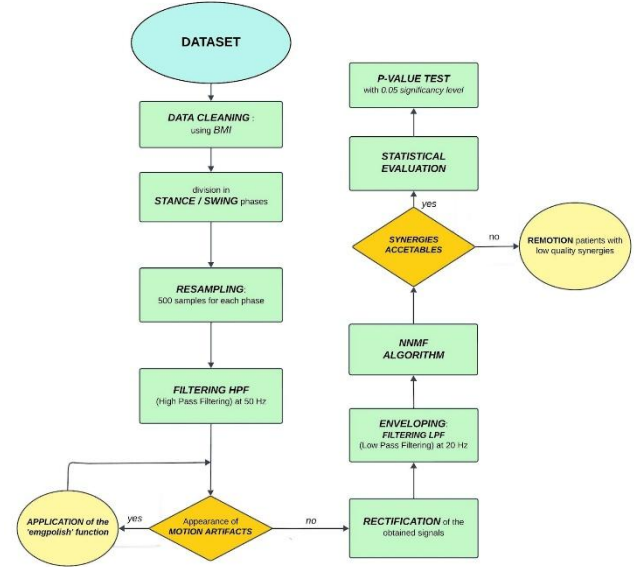


Figure 1. Flow-chart representing our work.

III. RESULTS

In the data cleaning phase, we evaluated the BMI coefficient as shown in (1) and we found out that the subject n° 28 was a woman outlier, so we decided to remove it. As shown in the boxplot image (Fig. 2), in which she is represented as a red dot, the subject was far away from the set of other BMI values.

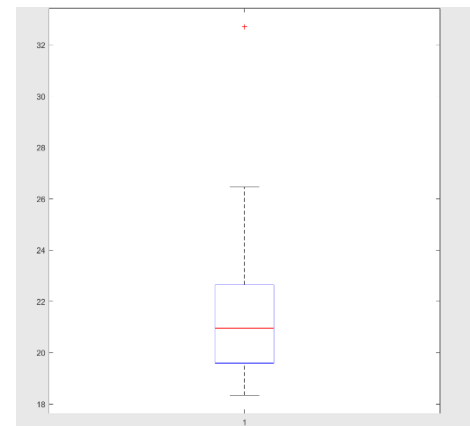


Figure 2. Boxplot of women subgroup.

Subsequently, after the resampling phase, signals were high pass filtered, and we found out that there were still many motion artifacts, as shown in Fig.3, present for several subjects with amplitudes often much higher than the normal physiological signal.



Figure 3. Motion artifacts on EMG signal from MA muscle.

The problem was that this type of artifact couldn't be removed with the high pass filtering due to its high frequency. For this reason, we used the function *emgpolish.m* previously described.

Once we got acceptable signals, we rectified and low pass filtered them, obtaining the envelope; then we applied the NMF algorithm to each subgroup. This algorithm was provided to us by the professors during the lectures.

We performed a sorting trial using a similarity matrix calculated thanks to the cross-correlation among the several time profiles, considering the order of the first one as the reference. This method worked well with the sorting of synergies related to the walking task but resulted in several misclassifications of synergies related to the running task, this created a bias in the final comparison. So, we opted for the Hungarian algorithm which gave us better performances also related to the running task [4].

For our work we found four as the ideal number of synergies. In figures from 4 to 7 it is represented the average trending of the synergies for the subgroups. Each synergy corresponds to a specific biomechanical function: weight acceptance (C1-W1); propulsion (C2-W2); early swing (C3-W3) and late swing (C4-W4).

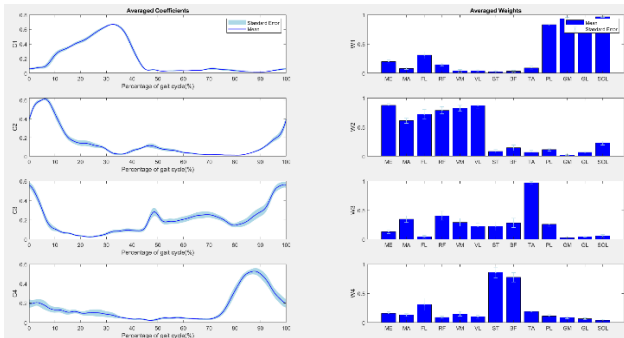


Figure 4. Men Walk - Average trend in synergies.

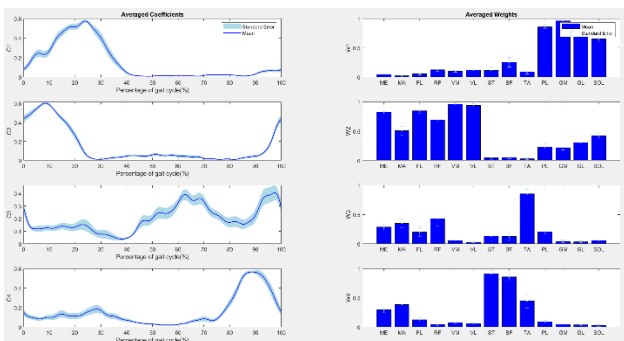


Figure 5. Men Run - Average trend in synergies.

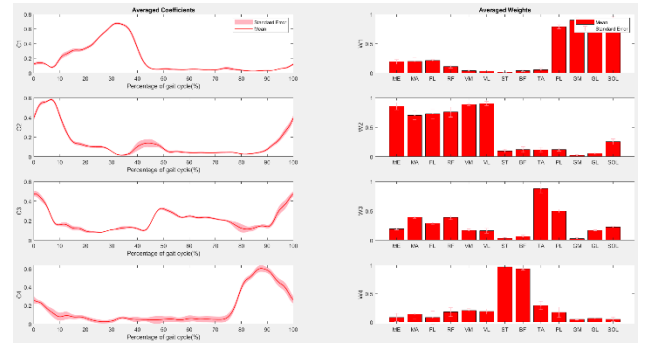


Figure 6. Women Walk - Average trend in synergies.

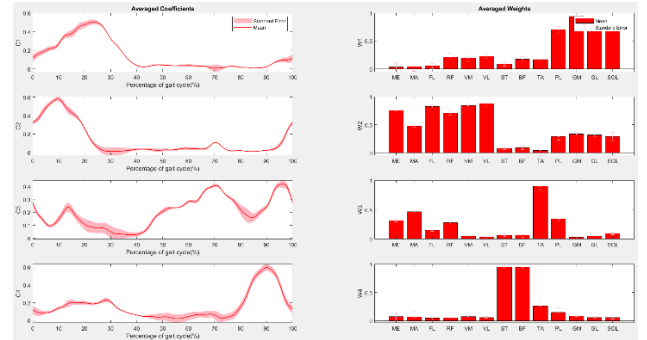


Figure 7. Women Run - Average trend in synergies.

Before obtaining this correct number of synergies, we removed subject n° 13, 19, 30 from the women subgroup and n° 21, 27 from the men one. We had to remove them since they were causing problems with synergies or they had EMG signals greatly corrupted by noise.

The last part of the work consisted of a statistical evaluation using the student's t-test [5]. We performed two types of comparisons: the first compared the average trend in synergies related to the same type of locomotion between the two genders; the second compared walking and running tasks between individuals of the same sex.

The results of the statistical tests obtained for the different subgroups are shown in Table 1,2 and 3.

Table 1. P-Value comparisons.

Comparisons	P-Values			
	C1	C2	C3	C4
MW-WW	6.696e-14	2.316e-01	1.114e-02	4.402e-13
MR-WR	3.680e-16	2.834e-74	9.296e-03	5.324e-02
MW-MR	0	2.130e-23	7.209e-01	2.267e-14
WW-WR	0	7.623e-14	3.575e-01	1.072e-02
Colorbar				
p-value >= 0.05		0.01 <= p-value < 0.05		p-value < 0.01

Table 2. P-value comparisons.

Muscles	Men Walk-Men Run: P Values			
	W1	W2	W3	W4
ME	6.036e-07	3.072e-01	1.275e-02	1.498e-02
MA	3.573e-03	2.254e-01	9.896e-01	1.203e-02
FL	7.338e-05	1.669e-01	6.300e-02	9.168e-02
RF	5.535e-01	3.286e-01	8.700e-01	1.152e-01
VM	2.857e-02	8.992e-03	6.400e-04	7.364e-02
VL	2.338e-02	1.554e-01	4.260e-04	2.008e-01
ST	9.819e-02	3.396e-01	3.615e-01	6.245e-01
BF	8.089e-03	9.960e-02	1.806e-01	4.338e-01
TA	9.135e-01	2.535e-01	1.290e-01	4.940e-02
PL	4.753e-01	5.822e-02	3.302e-01	5.666e-01
GM	4.455e-01	2.816e-06	6.443e-01	1.986e-01
GL	7.918e-01	4.810e-04	4.722e-01	1.759e-01
SL	4.702e-10	4.833e-03	7.734e-01	3.777e-01
Colorbar				
p-value ≥ 0.05		0.01 $\leq$ p-value < 0.05		p-value < 0.01

Table 3. P-value comparisons

Muscles	Women Walk-Women Run: P Values			
	W1	W2	W3	W4
ME	7.917e-05	3.250e-01	2.156e-01	8.415e-01
MA	2.810e-02	5.826e-02	5.896e-01	1.046e-02
FL	1.517e-06	3.118e-01	2.261e-01	2.896e-01
RF	3.335e-02	6.613e-01	2.887e-01	1.017e-03
VM	1.926e-02	4.573e-01	4.815e-03	1.093e-05
VL	1.779e-03	6.560e-01	7.544e-03	3.252e-03
ST	3.898e-03	8.073e-01	1.700e-01	3.898e-01
BF	4.002e-05	6.358e-01	8.691e-01	8.277e-01
TA	2.100e-01	1.420e-01	8.292e-01	7.434e-01
PL	2.529e-01	6.762e-03	1.957e-01	6.176e-01
GM	4.026e-01	1.995e-09	6.506e-01	3.888e-01
GL	3.355e-02	6.412e-05	1.416e-02	5.824e-01
SL	1.151e-02	5.660e-01	5.837e-03	7.260e-01
Colorbar				
p-value ≥ 0.05		0.01 $\leq$ p-value < 0.05		p-value < 0.01

IV. DISCUSSION

We divided the dataset into four subgroups because we expected differences in trends between men and women doing the same task and between running and walking among individuals of the same gender. For instance, we hypothesized that running would elicit greater activation of certain muscles compared to walking, as the biomechanical demands of running are inherently more intense.

During the course of our analysis, several methodological choices were made to improve the reliability of our results. We specifically chose to avoid the cross-correlation method, used in other works [1], and to instead perform the Hungarian algorithm [4] for sorting the synergies. In this case, it outperforms the cross-correlation method as it is particularly effective for optimally matching multiple signals. It is well-suited for scenarios involving more than two signals, where finding the best matches based on a cost matrix is required, making it ideal for multi-dimensional matching problems. EMG signals fall into this category, as they involve multiple muscles and are inherently multi-dimensional.

We expected to get four as the best number of synergies. This expectation was based on the knowledge that four is the number of the main biomechanical function, as we saw during the course lessons. Our work validated the hypothesis, because four resulted as the mode of the optimal synergies. This led us to the average trends showed in figures from 4 to 7.

We chose to apply statistical analysis using a p-value test [5], being this one of the most reliable methods for assessing the statistical relevance of the results obtained from the synergies study.

By examining the results presented in Tables from 1 to 3, we made the following considerations about our work:

1. Minimal Differences Between Genders

Contrary to our initial hypothesis, we found that the differences between male and female subgroups were less significant than expected. Statistical analysis revealed that muscle synergies showed minimal variation based on the sex of the participants. This may be attributed to the fact that men and women share a similar muscle activation process, governed by the same neural activation patterns. A possible explanation lies in the developmental origins of synergies, which are fully established in adulthood but are developed and refined during childhood, a period when a greater number of synergies are typically observed, as discussed extensively in article [6].

2. Significant Differences Between Running and Walking

In contrast to the gender comparison, the test showed instead a significant difference between the walking and running subgroups. We anticipated this difference, and our work and analysis confirmed this hypothesis. As you can see in the tables, the comparison between subgroup-3 and subgroup-4 yielded many significant results. Similarly, the comparison between subgroup-1 and subgroup-2, also revealed substantial differences.

This can be justified by the fact that running is a fundamentally different activity, involving different biomechanical demands compared to walking. For instance,

running also include a floating phase, which is absent in walking.

As it can be seen in the tables and in muscle contributions, the weight acceptance phase (C1-W1) is the one most prone to a statistical difference between walking and running. During this phase, power absorption occurs to decelerate the centre of mass of the body (i.e. brake), preventing joint collapse, maintaining forward propulsion and providing vertical support, as shown in article [7]. The main biomechanical function of this phase is to provide stability to the entire body, absorb impact and gradually transfer weight to the other limb. For this reason, muscles, and consequently also the synergies that activate them, are much more stressed and employed during running phase.

From these observations, it is evident that muscle activations differ significantly between walking and running. This conclusion is supported both by the theory of muscle activation and by the results of our synergies study.

In conclusion, our study confirmed that muscle synergies are significantly influenced by the type of activity performed by the subjects, such as walking or running. This finding highlights the substantial role that task-specific biomechanical and neural demands play in shaping muscle coordination patterns.

On the other hand, our initial hypothesis regarding differences in muscle synergies between male and female participants was not supported. The results showed no significant variation in muscle synergies based on sex, suggesting that men and women share similar underlying neural mechanisms for muscle activation. This indicates that the organization of muscle synergies is likely determined by universal neural and biomechanical principles, rather than by sex-specific factors.

REFERENCES

- [1] Saito, A., Tomita, A., Ando, R. et al., "Muscle synergies are consistent across level and uphill treadmill running", *Sci Rep* 8, 5979 (2018)
- [2] Santuz, Alessandro, et al. "Modular control of human movement during running: an open access data set." *Frontiers in physiology* 9 (2018): 1509.
- [3] Mileti, Ilaria, et al. "Muscle activation patterns are more constrained and regular in treadmill than in overground human locomotion." *Frontiers in Bioengineering and Biotechnology* 8 (2020).
- [4] Kunkun Zhao, Zhisheng Zhang, Haiying Wen, Zihan Wang, Jiankang Wu, "Modular Organization of Muscle Synergies to Achieve Movement Behaviors" *J Healthc Eng.* 2019 Nov 15. pp. 2-3.
- [5] Andrade C. "The P Value and Statistical Significance: Misunderstandings, Explanations, Challenges, and Alternatives." *Indian J Psychol Med.* 2019 May-Jun; 41(3):210-2.
- [6] Bach MM, Daffertshofer A and Dominici N (2021), "Muscle Synergies in Children Walking and Running on a Treadmill." *Front. Hum. Neurosci.*
- [7] Lise Worthen-Chaudhari, Julie Bing, James P. Schmiedeler, D. Michele Basso, "A new look at an old problem: Defining weight acceptance in human walking, *Gait & Posture*" Volume 39, Issue 1, 2014.